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MEDICAL INSURANCE PRICE PREDICTION

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ABSTRACT

The **Medical Insurance Price Prediction** project aims to enhance the accuracy and fairness of insurance cost predictions using advanced machine learning techniques. It analyzes critical factors such as age, health status, medical history, lifestyle choices, and location. By merging two datasets from Kaggle based on the common column 'age,' the study provides a comprehensive approach to risk assessment. Various machine learning models, including linear regression and gradient boosting, are tested to determine the optimal prediction method. The project seeks to overcome limitations in traditional insurance pricing methods by reducing bias and improving personalization. The expected outcomes will benefit both insurers and policyholders by ensuring more transparent and data-driven premium calculations. This initiative contributes to the advancement of predictive analytics in healthcare financing.

Keywords: Machine Learning.

I. INTRODUCTION

Medical insurance pricing plays a crucial role in managing healthcare costs, yet traditional actuarial methods often fail to account for complex risk factors. Factors such as age, medical history, lifestyle choices, and location significantly impact insurance premiums. Machine learning techniques offer a more precise and data-driven approach to predicting insurance costs. This project leverages a Kaggle dataset, integrating demographic and health-related data for improved accuracy. By applying models like linear regression and gradient boosting, it enhances fairness and personalization in premium calculations. The proposed system aims to optimize risk assessment, reduce pricing biases, and improve transparency in medical insurance.

II. METHODOLOGY

Architecture Overview:

The goal of Architecture Design (AD) is to give the internal design of the actual program code for the Insurance Premium Prediction. AD describes the class diagrams with the methods and relation between classes and program specification. It describes the modules so that the programmer can directly code the program from the document.

Data Flow:

Data Collection: The project uses a Kaggle dataset containing demographic details (age, gender, BMI), medical history (chronic diseases, smoking status), and policy attributes.

Data Preprocessing: Missing values are handled, categorical variables are encoded, and numerical features are scaled to ensure consistency.

Feature Selection & Engineering: Key predictive features are selected using techniques like Recursive Feature Elimination (RFE) and Shapley Additive Explanations (SHAP).

Model Training: Machine learning models, including Linear Regression, Decision Trees, Random Forest, and Gradient Boosting, are trained to identify the best-performing model.

Model Evaluation: Performance is assessed using metrics such as Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 score to ensure accuracy and reliability.

Deployment: The final model is deployed using Flask to create a user-friendly web application for real-time insurance cost predictions.

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III. MODELING AND ANALYSIS

Data Preparation: Collected demographic and medical data, handled missing values, and encoded categorical features.

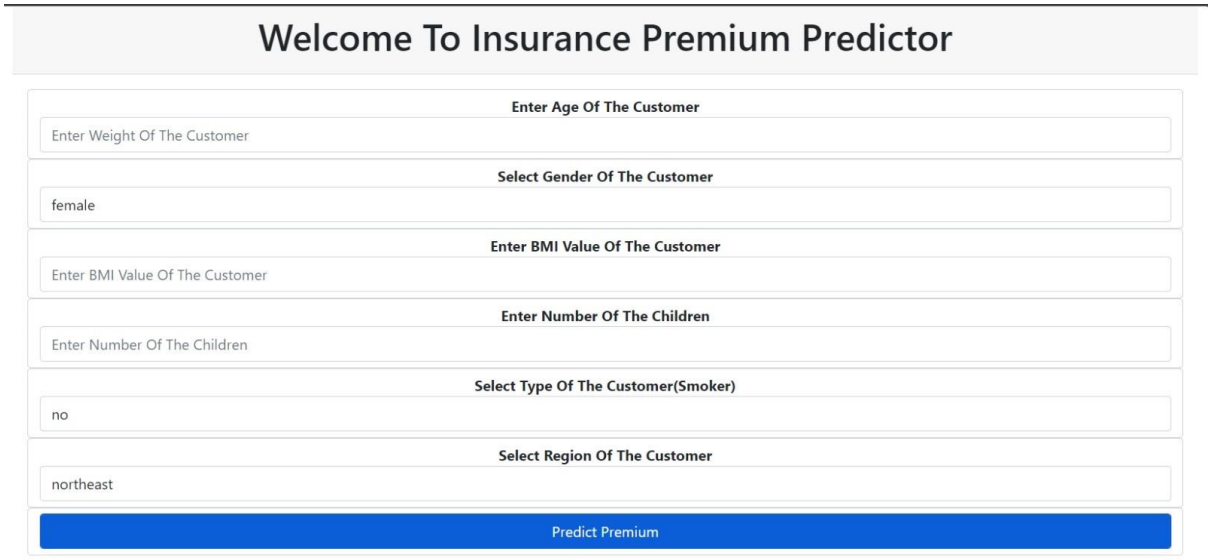
Feature Selection: Used RFE and SHAP to identify key predictors like age, BMI, and smoking status.

Model Training: Tested Linear Regression, Decision Trees, Random Forest, and Gradient Boosting, with Gradient Boosting performing best.

Evaluation: Assessed models using MSE, RMSE, and R^2, where Gradient Boosting achieved an R^2 of 0.8527.

Result Analysis: The model accurately predicts premiums, ensuring fair and personalized pricing.

Deployment: Integrated into a Flask web app for real-time predictions.



IV. RESULTS AND DISCUSSION

The Gradient Boosting model outperformed other machine learning models with an R^2 score of 0.8527 and the lowest RMSE of 4652.33, ensuring high prediction accuracy. The model effectively identifies key risk factors such as age, BMI, and smoking status, leading to fair and personalized insurance pricing. Compared to traditional actuarial methods, it provides better risk assessment and premium estimation. The system is deployed via a Flask web application for real-time insurance cost predictions.

V. CONCLUSION

The Medical Insurance Price Prediction system enhances accuracy, fairness, and transparency in premium calculations using machine learning. The Gradient Boosting model proved to be the most effective, outperforming traditional actuarial methods. By integrating key risk factors, it ensures personalized and data-driven insurance pricing. The deployed Flask web application allows real-time premium predictions, benefiting both insurers and policyholders.

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